FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
CONCEPT RECAPITULATION TEST – II
PAPER-1

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**CD**	19.	**BCD**	37.	**BCD**
2.	**BD**	20.	**ABC**	38.	**AC**
3.	**B**	21.	**AB**	39.	**AC**
4.	**BC**	22.	**ABC**	40.	**BD**
5.	**ACD**	23.	**ABD**	41.	**ACD**
6.	**D**	24.	**ACD**	42.	**AD**
7.	**BD**	25.	**BCD**	43.	**AC**
8.	**C**	26.	**B**	44.	**C**
9.	**B**	27.	**A**	45.	**D**
10.	**D**	28.	**A**	46.	**B**
11.	**C**	29.	**C**	47.	**B**
12.	**B**	30.	**B**	48.	**D**
13.	**A**	31.	**D**	49.	**A**
14.	**8**	32.	**5**	50.	**2**
15.	**5**	33.	**1**	51.	**1**
16.	**2**	34.	**8**	52.	**3**
17.	**2**	35.	**6**	53.	**5**
18.	**1**	36.	**5**	54.	**7**

Physics

PART – I

SECTION – A

1. CD

Sol. Particle *P* will have a velocity in vertical direction.

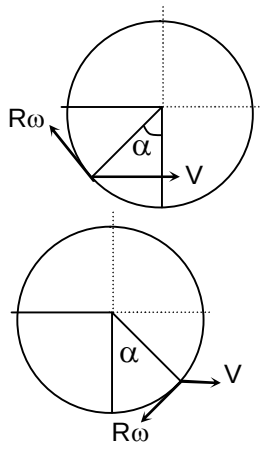
i.e. $R\,\omega\cos\alpha = V$

$$\alpha = \cos^{-1}\frac{V}{R\omega}$$

Required angle θ is $\theta = \pi - \alpha = \pi - \cos^{-1}\frac{V}{R\omega}$

And in second case $R\omega\cos\alpha = V$

$$\alpha = \cos^{-1}\frac{V}{R\omega},\ \theta = \pi + \cos^{-1}\frac{V}{R\omega}$$



2. BD

Sol. $\beta = \lambda\frac{D}{d}$

3. B

Sol. Apply KVL.

4. BC

Sol. $v_L = v_C = v_R$;

$\Rightarrow x_L = x_C = R$

when inductor is short circuited

$Z = \sqrt{R^2 + x_C^2} = \sqrt{2}\,R$

$\therefore\ I = \frac{30}{Z} = \frac{30}{\sqrt{2}R}$

$\therefore V_L = ix_L = \frac{30}{\sqrt{2}R} \times R = \frac{30}{\sqrt{2}}$

$\therefore$ (A) is incorrect and with similar calculations (B) will be correct.

Here f_0 is the resonance frequency as $v_L = v_C$

$\omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}}$

$\frac{x_L}{x_C} = \frac{\omega L}{1/\omega C} = \omega^2 LC$

Given $f = 2f_0$

$\Rightarrow\ \omega = 2\omega_0$

$\therefore\ \frac{x_L}{x_C} = 4$

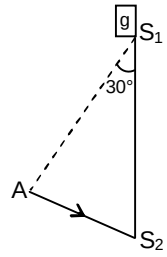
$\therefore$ (C) is also correct.

5. ACD

Sol. Apply Gauss law and property of conductor.

6. D

Sol. Extra phase change in glass = phae change in water of length AS_2.



$$\frac{2\pi t}{\lambda g}-\frac{2\pi}{\lambda\omega}t=\frac{2\pi}{\lambda\omega}\cdot\frac{2}{3}\sin 30^o$$

$$\Rightarrow \quad t\left[\frac{3}{2}-\frac{4}{3}\right]=\frac{4}{3}\times\frac{1}{3}\Rightarrow t=\frac{8}{3}\text{mm}.$$

7. BD

Sol. $E\propto\frac{1}{n^2}$ and total energy is –ve.

8. C
9. B
10. D

(Sol. for 8 – 10) Convert into polytropic process and use formulas of C and ΔW.

11. C
12. B
13. A

(Sol. for 11 – 13) Apply principle of caloriemetry.

SECTION – C

14. 8

Sol $F.x-\mu m_1gx-\frac{1}{2}kx^2=0$

$kx=\mu m_2g$ for just shifting m_2

$$F.x-\mu m_1gx-\frac{1}{2}\mu m_2gx=0$$

$$F=\mu\left(m_1+\frac{m_2}{2}\right)g=0.4\left(1+\frac{2}{2}\right)(10)\ =8\text{N}$$

15. 5

Sol. At stable equilibrium, U is minimum.

$\frac{dU}{dx}=0$ and $\frac{d^2U}{dx^2}>0$

$$=\frac{1}{dx}\left(\frac{x^3}{3}-\frac{ax^2}{2}+20x\right)=0.$$

$\Rightarrow \quad x^2-9x+20=0. \quad \Rightarrow \quad (x-5)(x-4)=0.$

$x = 5$ and $x = 4$ are points of equilibrium.

And U minimum when $\frac{d^2U}{dx^2}>0$. i.e. at $x = 5$.

16. 2

Sol. $\delta = x + y - z$

17. 2

Sol. $T = 2\pi\sqrt{\frac{1}{g}} \Rightarrow T^2 = 4\pi^2 1/g \Rightarrow g = \frac{4\pi^2 l}{T^2}$

Here % error in l = $\frac{1mm}{100cm} \times 100 = \frac{0.1}{100} \times 100 = 0.1\%$

and % error in T = $\frac{0.1}{2 \times 100} \times 100 = 0.05\%$

$\therefore$ % error in g = % error in l + 2 (% error in T) = $0.1 + 2 \times 0.05 = 0.2\%$.

18. 1

Sol. 30 VSD = 29 MSD

1 VSD = $\frac{29}{30}$MSD

L.C. = 1 MSD – 1 VSD

= $\left(1 - \frac{29}{30}\right)MSD = \frac{1}{30} \times 0.5^0 = 1$ minute

Chemistry

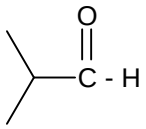
PART – II

SECTION – A

19. BCD
Sol. These concentrated terms don't depend upon volume.

20. ABC
Sol. $LiAlH_4$, converts carbonyl compound into alcohol.

21. AB
Sol. In peroxy acid anti-addition takes place.

22. ABC
Sol.  undergoes aldol condensation.

23. ABD
Sol. $R - \overset{\overset{O}{\|}}{C} - NH_2$ has least +ve charge density among all of the given compound.

24. ACD
Sol. It is benzoin condensation.

25. BCD
Sol. HOBr is Br_2/H_2O.

26. B
Sol. Liberman's test is used for phenolic group.

27. A
Sol. Dye test is used for aromatic amine.

28. A
Sol. Ring test is for NO_3^- ion.

29. C
Sol. Saponification is second order kinetics.

30. B
Sol. Ester hydrolysis in acidic medium is first order kinetics.

31. D
Sol. It is first order kinetics.

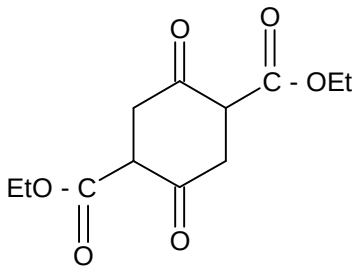
SECTION – C

32. 5
Sol. Chiral C is attached to 4 different groups.

33. 1
Sol. Conc. H_2SO_4 is a dehydrating agent.

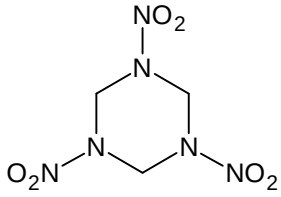
34. 8
Sol.

A =



35. 6

Sol. Compound (Z) is



36. 5
Sol. Fructose is a ketose sugar.

Mathematics

PART – III

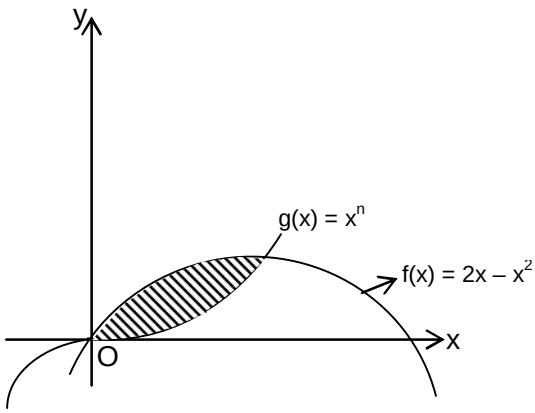
SECTION – A

37. BCD

Sol. Solving $f(x) = 2x - x^2$ and $g(x) = x^n$

We have $2x - x^2 = x^n \Rightarrow x = 0$ and $x = 1$

$$A = \int_0^1 \left(2x - x^2 - x^n\right) dx = x^2 - \frac{x^3}{3} - \frac{x^{n+1}}{n+1}\Big]_0^1$$



$$= 1 - \frac{1}{3} - \frac{1}{n+1} = \frac{2}{3} - \frac{1}{n+1}$$

Hence, $\frac{2}{3} - \frac{1}{n+1} = \frac{1}{2} \Rightarrow \frac{2}{3} - \frac{1}{2} = \frac{1}{n+1}$

$$\Rightarrow \frac{4-3}{6} = \frac{1}{n+1} \Rightarrow n+1 = 6 \Rightarrow n = 5$$

Hence n is a divisor of 15, 20, 30

38. AC

Sol. $f(x) = 2x^4 + x^4 \sin\frac{1}{x}$ $\quad x \neq 0$

$= 0$ $\quad x = 0$

$$f'(x) = 8x^3 + 4x^3 \sin\frac{1}{x} - x^2 \cos\frac{1}{x}$$

39. AC

Sol. $\sqrt{(x-5)^2 + (y-7)^2} + \sqrt{(x+1)^2 + (y+1)^2} = 12$

$SP + S'P = 2a \Rightarrow$ conic is ellipse

$2ae = \sqrt{36+64}$

$2a = 12 \Rightarrow e = \frac{5}{6}$

Major axis $y - 7 = \frac{7+1}{5+1}(x-5)$

$\Rightarrow 4x - 3y + 1 = 0$

40. BD

Sol. All roots of equation $ax^4 + bx^2 + c = 0$ will be real if both roots of $ay^2 + by + c = 0$ will be positive (let $x^2 = y$)

i.e. sum of roots $= -\frac{b}{a} > 0$ and product of roots $= \frac{c}{a} > 0$.

Hence, a and b are of opposite sign, while a and c are of same sign.

41. ACD

Sol. Let common difference is α

(A) $a(a+3\alpha) < (a+\alpha)(a+2\alpha)$

$a^2 - 3a\alpha < a^2 + 3a\alpha + 2\alpha^2$

$\Rightarrow 2\alpha^2 > 0,$ which is true.

(B) $a + a + 2\alpha < a + \alpha + a + 3\alpha$

$0 < 2\alpha,$ false is $a > b > c > d > 0$

(C) $a + a + 3\alpha = a + \alpha + a + 2\alpha$

(D) $(a+1),(b+1),(c+1),(d+1)$ all in A.P.

$\Rightarrow (a+1)(d+1) < (b+1)(c+1)$

42. AD

Sol. $\because 3^{4n} = 81^n = (1+80)^n = 1+80\lambda,\ \lambda \in N$

$\therefore 3^{3^{4n}} = 3^{1+80\lambda} = 3.3^{80\lambda}$

$= 3.(9)^{40\lambda} = 3(10-1)^{40\lambda}$

$= 3(1+10\mu) = 3 + 30\mu$

$\therefore$ Last digit of $3^{3^{4n}} + 1$ is 4.

43. AC

Sol. Suppose the two players did not play at all so that the remaining (n – 2) players played ${}^{n-2}C_2$ matches. Since, these two players played 3 matches each, hence the total number of matches is ${}^{n-2}C_2 + 3 + 3 = 84$ (given)

$\Rightarrow {}^{n-2}C_2 = 78 = {}^{13}C_2$

$\Rightarrow n - 2 = 13$

$\therefore n = 15 = {}^6C_2$

44. C

45. D

46. B

(Sol. 44 to 46)

(I) For domain of $f(x) = \ell n\left(x^2 - 6x + 5\right)$

$x^2 - 6x + 5 > 0$

$(x-5)(x-1) > 0$

$x > 5$ or $x < 1$

(II) $y = \frac{\{x\}}{1+\{x\}}$.

$\Rightarrow \{x\} = \dfrac{y}{1-y}$. Hence $0 \le \dfrac{y}{1-y} < 1$

(III) Given, $f(x) = \sin\left(2\pi\left(\dfrac{\sqrt{x+7}-4}{x-9}\right)\right)$

domain of $f(x) = [-7, \infty) - \{9\}$

$g(x) = \dfrac{\pi(x+7-16)}{(x-9)\left(\sqrt{x+7}+4\right)}$ (Rationalise)

$= \dfrac{\pi}{\sqrt{x+7}+4}$

So, range of $g(x)$ is $\left(0, \dfrac{\pi}{4}\right] - \left\{\dfrac{\pi}{8}\right\}$.

Hence, range of $f(x)$ is $(0,1] - \left\{\dfrac{1}{\sqrt{2}}\right\}$.

(IV) Let $x^2 = 4\cos^2\theta + \sin^2\theta$ then $\left(4 - x^2\right) = 3\sin^2\theta$ and $\left(x^2 - 1\right) = 3\cos^2\theta$

$\therefore \quad f(x) = \sqrt{3}|\sin\theta| + \sqrt{3}|\cos\theta|$

$\Rightarrow \quad y_{min} = \sqrt{3}$ and y_{max}

$= \sqrt{3}\left(\dfrac{1}{\sqrt{2}} + \dfrac{1}{\sqrt{2}}\right) = \sqrt{6}$

Hence range of $f(x)$ is $\left[\sqrt{3}, \sqrt{6}\right]$

47. B
48. D
49. A

(Sol. 47 to 49)

(I) Let P (x) = 6 + x, for x = 0, 1, 2, 3

Let $f(x) = P(x) - x - 6$

$\Rightarrow f(x)$ is also a polynomial of degree 4.

$\Rightarrow f(x) = x(x-1)(x-2)(x-3)$

As $f(x)$ vanishes at x = 0, 1, 2, 3, so $f(x) = P(x) - x - 6 \equiv (x-0)(x-1)(x-2)(x-3)$

$\Rightarrow P(x) = x + 6 + x(x-1)(x-2)(x-3)$

Hence $P(4) = 10 + (4.3.2.1) = 10 + 24 = 34$

(II) $\displaystyle\sum_{n=1}^{\infty} \tan^{-1}\left(\frac{2}{n^2+n+4}\right) = \sum_{n=1}^{\infty} \tan^{-1}\left\{\frac{\dfrac{2}{n(n+1)}}{1+\dfrac{4}{n(n+1)}}\right\}$

$= \displaystyle\sum_{n=1}^{\infty} \tan^{-1}\left\{\frac{\dfrac{2}{n} - \dfrac{2}{n+1}}{1+\dfrac{2}{n}.\dfrac{2}{n+1}}\right\}$

$$= \sum_{n=1}^{\infty}\left\{\tan^{-1}\left(\frac{2}{n}\right) - \tan^{-1}\left(\frac{2}{n+1}\right)\right\}$$

$$= \left\{\tan^{-1}\left(\frac{2}{1}\right) - \tan^{-1}\left(\frac{2}{2}\right)\right\} + \left\{\tan^{-1}\left(\frac{2}{2}\right) - \tan^{-1}\left(\frac{2}{3}\right)\right\} + ...\infty \text{ terms}$$

$$= \tan^{-1}2 = \tan^{-1}\left(\frac{a}{b}\right)$$

$$\Rightarrow a = 2;\ b = 1 \ \Rightarrow (a+b) = 3$$

(III) $P = \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \frac{1}{5^4} + \frac{1}{6^4}.....\infty$

$$P = \left(\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + ...\right) + \frac{1}{2^4}\left(\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + ...\right)$$

$$P = q + \frac{1}{16}p$$

$$\frac{15}{16}p = q \ \Rightarrow \frac{p}{q} = \frac{16}{15}$$

(IV) Given $s_1 + s_2 + s_3 + ... + s_{10} = t_1 + t_2 + t_3 + ... + t_{15}$ Let 1st sequence is $a_1, a_1 + d_1, a_1 + 2d_1, ...$ and 2nd is $a_1, a_1 + d_2, a_1 + 2d_2.....$ (since $s_1 = t_1$)

Given $s_2 = 2t_2$

$\therefore \quad a_1 + d_1 = 2(a_1 + d_2)$

$\therefore \quad a_1 = d_1 - 2d_2$ ……(i)

We have to find $\frac{s_2 - s_1}{t_2 - t_1} = \frac{d_1}{d_2} = ?$

Now $\frac{10}{2}[2a_1 + 9d_1] = \frac{15}{2}[2a_1 + 14d_2]$

This gives $a_1 = 9d_1 - 21d_2$ ……(ii)

From equation (i) and (ii) $\frac{d_1}{d_2} = \frac{19}{8}$

SECTION – C

50. 2

Sol. Since $x(1-x) = \frac{1}{4} - \left(x - \frac{1}{2}\right)^2$, so put $x = \frac{1}{2} + \left(\frac{1}{2}\right)\sin\theta$.

Therefore, $\int_0^1 \frac{dx}{(1+x)(2+x)\sqrt{x(1-x)}}$

$$= \int_{-\pi/2}^{\pi/2} \frac{\left(\frac{1}{2}\right)\cos\theta\, d\theta}{\left(\frac{1}{4}(1+\sin\theta)^2 + \frac{3}{2}(1+\sin\theta) + 2\right)\frac{1}{2}\cos\theta}$$

$= 4\int_0^{\pi} \frac{d\phi}{\cos^2\phi - 8\cos\phi + 15}$ where, $\phi = \theta + \frac{\pi}{2}$

$= \frac{4}{2}\left[\int_0^{\pi} \frac{d\phi}{\cos\phi - (4+1)} - \int_0^{\pi} \frac{d\phi}{\cos\phi - (4-1)}\right]$

Let $I_1 = \int_0^{\pi} \frac{d\phi}{\cos\phi - a}$ where $a > 1$

Put $\tan\left(\frac{\phi}{2}\right) = t$, $d\phi = 2\frac{dt}{(1+t^2)}$.

$\therefore I_1 = \int_0^{\infty} \frac{\frac{2dt}{(1+t^2)}}{\frac{(1-t^2)}{(1+t^2)-a}} = \frac{-2}{1+a}\int_0^{\infty} \frac{dt}{r^2 + \frac{(a-1)}{(a+1)}}$

$= \frac{-2}{1+a}\sqrt{\frac{a+1}{a-1}}\left(\tan^{-1}\frac{t\sqrt{a+1}}{\sqrt{a-1}}\right)\Bigg]_0^{\infty}$

$= \frac{-2}{\sqrt{a^2-1}}\cdot\frac{\pi}{2} = \frac{-\pi}{\sqrt{a^2-1}}$

Thus $I = 2\left[-\frac{\pi}{\sqrt{24}} + \frac{\pi}{\sqrt{8}}\right] = \pi\left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{6}}\right) = \pi\frac{(\sqrt{3}-1)}{\sqrt{6}}$

51. 1

Sol. Let $y = \frac{e^x - e^{-x}}{2} \Rightarrow e^{2x} - 1 = 2ye^x$

Therefore, $t^2 - 2yt - 1 = 0$, $t = e^x$

$\Rightarrow t = \frac{2y \pm \sqrt{4y^2+4}}{2} = y \pm \sqrt{y^2+1}$

$\Rightarrow x = \log\left(y + \sqrt{y^2+1}\right)$ (since $e^x > 0$)

$\therefore f^{-1}(x) = g(x) = \log\left(x + \sqrt{x^2+1}\right)$

$g\left(\frac{e^{1002}-1}{2e^{501}}\right) = \log\left(\frac{e^{1002}-1}{2e^{501}} + \frac{e^{1002}+1}{2e^{501}}\right) = \log e^{501} = 501$

52. 3

Sol. Taking logarithm of both sides, we get $\log y = x\left[\log\left(1+\frac{1}{x}\right)\right]$

$\Rightarrow \frac{1}{y}y_1(x) = \frac{x^2}{x+1}\left(-\frac{1}{x^2}\right) + \log\left(1+\frac{1}{x}\right)$

$= -\frac{1}{x+1} + \log\left(1+\frac{1}{x}\right)$ (i)

Since $y(2)=\left(1+\frac{1}{2}\right)^2=\frac{9}{4}$ so $y_1(2)=\left(\frac{9}{4}\right)\left(-\frac{1}{3}+\log\frac{3}{2}\right)$

Multiplying (i) by y and then differentiating, we get

$$y_2(x)=y_1(x)\left(-\frac{1}{x+1}+\log\left(1+\frac{1}{x}\right)\right)+y(x)\times\left(\frac{1}{(x+1)^2}+\frac{x}{x+1}\left(-\frac{1}{x^2}\right)\right)$$

So $y_2(2)=y_1(2)\left(-\frac{1}{3}+\log\frac{3}{2}\right)+y(2)\left(\frac{1}{9}-\frac{1}{6}\right)=\left(\frac{9}{4}\right)\left(-\frac{1}{3}+\log\frac{3}{2}\right)^2-\frac{1}{8}$.

53. 5

Sol. The probability of getting at least one head
= 1 – probability of getting no heads

$$=1-{}^nC_0\left(\frac{1}{2}\right)^0\cdot\left(\frac{1}{2}\right)^n=1-\frac{1}{2^n}$$

The probability of getting at least two tails
= 1 – probability of getting no tails – probability of getting 2 tails

$$=1-{}^nC_n\left(\frac{1}{2}\right)^n\cdot\left(\frac{1}{2}\right)^0-{}^nC_{n-1}\left(\frac{1}{2}\right)^{n-1}\cdot\frac{1}{2}=1-\frac{1}{2^n}-n\frac{1}{2^n}$$

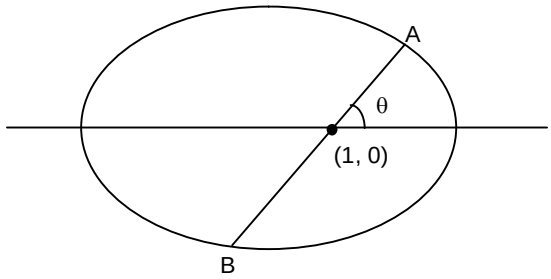
From the question,

$$\left(1-\frac{1}{2^n}\right)-\left(1-\frac{1+n}{2^n}\right)=\frac{5}{32}$$

or $\frac{n+1}{2^n}-\frac{1}{2^n}=\frac{5}{32}$ or $\frac{n}{2^n}=\frac{5}{32}\Rightarrow n=5$

54. 7

Sol. Let $AB=l_1$



Equation of $AB=\frac{x-1}{\cos\theta}=\frac{y-0}{\sin\theta}=r$

$$\Rightarrow\ \frac{(1+r\cos\theta)^2}{4}+\frac{(r\sin\theta)^2}{3}=1$$

$$\Rightarrow\ r^2\left(3+\sin^2\theta\right)+6r\cos\theta-9=0$$

$$\Rightarrow\ r_1-r_2=-\frac{6\cos\theta}{3+\sin^2\theta},r_1r_2=\frac{9}{3+\sin^2\theta}$$

$$l_1=|r_1|+|r_2|=\sqrt{(r_1-r_2)^2+4r_1r_2}=\frac{12}{3+\sin^2\theta}$$

$$l_2=\frac{12}{3+\sin^2\left(\frac{\pi}{2}+\theta\right)}=\frac{12}{3+\cos^2\theta}$$

$$\Rightarrow\frac{1}{l_1}+\frac{1}{l_2}=\frac{7}{12}$$